BUSINESS SCIENCE STUDY PLAN

signal -> meaning -> value -> execution

Goal: master the science of business by learning to observe business signals, interpret meaning, create measurable value, execute decisions, and improve the system through feedback.

One-line definition: business science is the discipline of converting observed business reality into tested decisions that improve cash, demand, efficiency, compliance, resilience, and strategic advantage.

Executive Overview

The 13 subjects combine into one operating system: System Science defines the model, Operations Management creates activity, AIS records activity, Financial Accounting reports it, Financial Statement Analysis interprets it, Managerial and Cost Accounting control it, Corporate Finance allocates capital to it, Taxation changes after-tax cash, Auditing tests trust, Business Law defines enforceable boundaries, Marketing creates demand, and Music Business provides a concrete industry application.

Core logic: signal is observed reality; meaning is interpretation; value is the business result; execution is the action and feedback loop.

Source Index Method

Root scanned: /Users/charlesdaniel/Documents/Business Science . Indexed source files: 457 usable files across all required subfolders. The full source index is exported as BUSINESS_SCIENCE_SOURCE_INDEX.csv.

Extraction rule: every file was indexed by path, folder, extension, size, concept signals, and cluster. Folder-provided Markdown, CSV, TSV, JSON, and text manifests were read directly as the high-signal content layer. PDFs were cataloged from stable file/path metadata and folder manifests because the corpus has hundreds of large PDFs.

Folder Inventory

Subfolder	Files	Main concepts	Business function	Contribution
Accounting Information Systems	21	systems thinking; AIS and ERP; accounting cycle; data quality; optimization	Turns operational events into reliable, structured business data.	Turns activity into structured, controlled, auditable data.
Auditing	36	audit and assurance; data quality; risk and return; systems thinking; process design	Tests evidence, controls, risk, and trust in reported information.	Tests whether signals, controls, statements, and claims are trustworthy.
Business Law	82	legal structure; tax compliance; process design; systems thinking; ratio analysis	Defines the enforceable rules, rights, governance structures, and compliance boundaries.	Sets enforceable boundaries, incentives, rights, duties, and remedies.
Corporate Finance	48	capital allocation; capital structure; risk and return; marketing demand; cash flow	Allocates capital across projects, risks, financing choices, and payouts.	Turns financial meaning into capital allocation, financing, and valuation decisions.
Cost Accounting	17	cost behavior; management control; process design; systems thinking; optimization	Measures resource consumption, cost drivers, unit economics, and process economics.	Turns resource use into unit economics, margin insight, and improvement targets.
Financial Accounting	20	financial reporting; systems thinking; legal structure; optimization; audit and assurance	Converts transactions into standardized financial statements.	Turns transactions into standardized external financial information.
Financial Statement Analysis	34	financial reporting; ratio analysis; capital allocation; capital structure; systems thinking	Interprets statements into profitability, liquidity, solvency, quality, and valuation signals.	Turns statements into diagnostic and forecasting meaning.
Managerial Accounting	24	management control; systems thinking; cost behavior; process design; optimization	Builds internal decision systems for planning, control, pricing, budgeting, and performance.	Turns internal data into planning, budgeting, and control decisions.
Marketing	22	marketing demand; systems thinking; capital structure; optimization; music revenue	Detects demand signals and designs revenue-generating customer systems.	Senses demand and converts customer behavior into revenue hypotheses.
Music Business	40	music revenue; marketing demand; systems thinking	Applies business science to rights, royalties, audiences, catalogs, artists, and creative assets.	Applies the full system to rights, royalties, audiences, platforms, and creative assets.
Operations Management	26	process design; systems thinking; optimization; legal structure; risk and return	Designs the activity engine that produces goods, services, costs, quality, and delivery outcomes.	Creates business activity and exposes flow, capacity, quality, and waste signals.
System Science	28	systems thinking; optimization; process design; legal structure	Provides the meta-model: feedback, stocks, flows, constraints, leverage, and system improvement.	Provides the meta-model for feedback, constraints, leverage, and improvement.
Taxation	59	tax compliance; risk and return; legal structure; optimization; systems thinking	Optimizes entity, transaction, timing, compliance, and cash-flow consequences under tax rules.	Turns legal and accounting classifications into after-tax cash consequences.

Master System Map

System flow: System Science -> Operations Management -> Accounting Information Systems -> Financial Accounting -> Financial Statement Analysis -> Managerial Accounting -> Cost Accounting -> Corporate Finance -> Taxation -> Auditing -> Business Law -> Marketing -> Music Business

- A business creates activity through customer demand, legal rights, labor, processes, assets, and contracts.
- Activity becomes data when events are captured in documents, systems, ledgers, and dashboards.
- Data becomes financial information when accounting rules classify, measure, summarize, and report it.
- Financial information becomes decisions when analysis converts statements, costs, cash, and risk into meaning.
- Decisions become strategy when capital, tax, legal, and market choices are prioritized against constraints.
- Strategy becomes execution through workflows, controls, budgets, campaigns, investments, and operating rules.
- Execution produces feedback through cash, demand, variance, audit evidence, legal outcomes, and customer behavior.

Study Plan Structure

Phase	Primary folders	System objective
Phase 1: Business as a System	System Science; Operations Management; Business Law	See business as activity inside rules and feedback.
Phase 2: Business Activity into Data	Accounting Information Systems; Financial Accounting; Auditing	Turn events into reliable, testable financial data.
Phase 3: Data into Financial Meaning	Financial Statement Analysis; Managerial Accounting; Cost Accounting	Turn records into diagnosis, control, and unit economics.
Phase 4: Financial Meaning into Capital Decisions	Corporate Finance; Taxation; Business Law	Turn analysis into after-tax value and structure decisions.
Phase 5: Market Demand and Revenue Systems	Marketing; Music Business; Operations Management	Turn demand signals into revenue systems and fulfillment discipline.
Phase 6: Integrated Business Science Labs	All folders	Operate the full observe -> decide -> execute -> improve loop.

Lesson Format

Each lesson includes lesson number, title, primary folders, specific files, objective, key concepts, signal, meaning, value, execution, exercise, mini case, deliverable, and questions.

Phase 1: Business as a System

Lesson 1: Business Science Operating Grammar

Primary folders used: System Science; Operations Management; Business Law

Core objective: Define business as a designed system that converts observed activity into controlled value creation.

Key concepts: stocks and flows; feedback loops; constraints; legal boundaries; operating model

Specific files used

- System Science/ INDEX.md
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ meadows_ leverage_ points.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ feedback.pdf
- Operations Management/ Operations_ Management_ PDF_ Index.md
- Business Law/ business_ law_ pdf_ index.md

Signal: Customer requests, contracts, work orders, delays, defects, costs, control failures, and market responses.

Meaning: Separate events from structure: identify stocks, flows, rules, incentives, constraints, and feedback loops.

Value: A shared model for diagnosing profit, risk, compliance, and execution problems without treating them as isolated issues.

Execution: Build a one-page business system map with inputs, transformations, outputs, constraints, metrics, and feedback.

Practical exercise: Map a small business from demand signal to fulfilled order to cash receipt; mark each feedback loop.

Mini case study: A service firm grows revenue but complaints and rework rise faster than sales. Diagnose whether the root is demand quality, capacity, process design, or control failure.

Deliverable: Business system map v1 with one measurable feedback loop.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define stocks and flows and feedback loops in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 2: Operations Activity Engine

Primary folders used: Operations Management; System Science; Business Law

Core objective: Translate business activity into process, capacity, flow, WIP, and control signals.

Key concepts: process map; Little's Law; capacity; value stream; compliance constraint

Specific files used

- Operations Management/ 01_ Seneca_ Introduction_ to_ Operations_ Management_ Process_ Capacity.pdf
- Operations Management/ 10_ Littles_ Law_ 50th_ Anniversary_ Flow_ Science.pdf
- Operations Management/ 21_ EPA_ Lean_ and_ Clean_ Value_ Stream_ Mapping.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ behavior.pdf
- Business Law/ 05_ Scientific_ Empirical_ Law/ formal- methods- business- process- compliance- systematic- review.pdf

Signal: Cycle time, lead time, WIP, queue length, utilization, rework, defects, handoffs, and policy violations.

Meaning: Find the constraint and explain how it shapes throughput, service quality, cost, and risk.

Value: Shorter lead time, lower waste, better asset use, stronger compliance, and more predictable delivery.

Execution: Create a value stream map and define one bottleneck-protection rule.

Practical exercise: Use WIP, throughput, and lead time to estimate whether a process problem is demand, capacity, or flow control.

Mini case study: A custom-order company has high utilization but missed delivery dates. Use flow metrics to show why local efficiency can reduce system performance.

Deliverable: Bottleneck diagnosis worksheet with one proposed operating rule.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define process map and Little's Law in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 3: Legal Structure as System Boundary

Primary folders used: Business Law; Taxation; Corporate Finance

Core objective: Explain how legal form, governance, tax identity, and agency incentives shape business behavior.

Key concepts: entity form; asset partitioning; agency cost; governance; tax classification

Specific files used

- Business Law/ 01_ Core_ Textbooks/ business- law- and- the- legal- environment- saylor.pdf
- Business Law/ 03_ Governance_ Legal_ Structure/ essential- role- of- organizational- law- hansmann- kraakman.pdf
- Business Law/ 03_ Governance_ Legal_ Structure/ g20- oecd- principles- of- corporate- governance- 2023.pdf
- Taxation/ IRS_ Pub_ 3402_ Taxation_ of_ LLCs.pdf
- Corporate Finance/ 32_ Jensen_ Meckling_ Theory_ of_ the_ Firm_ Agency_ Costs_ 1976.pdf

Signal: Formation documents, ownership rights, tax elections, board duties, contracts, and control authority.

Meaning: Interpret who controls assets, who bears risk, who receives returns, and who can bind the organization.

Value: Lower liability exposure, clearer incentives, improved financing access, and better compliance discipline.

Execution: Choose a legal/tax structure for a simple venture and state the control, tax, and capital consequences.

Practical exercise: Compare sole proprietorship, LLC, S corporation, and C corporation for the same business model.

Mini case study: Two partners start a music services firm. One funds equipment, one brings clients. Design the structure that protects assets and reduces incentive conflict.

Deliverable: Entity decision memo with risk, tax, governance, and finance tradeoffs.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define entity form and asset partitioning in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 4: Decision Models and System Design

Primary folders used: System Science; Operations Management; Managerial Accounting

Core objective: Use decision models to make assumptions explicit before committing resources.

Key concepts: requirements; optimization; decision rule; measurement system; model validation

Specific files used

- System Science/ 03_ System_ Design_ and_ Engineering/ nasa_ systems_ engineering_ handbook.pdf
- System Science/ 04_ Business_ Optimization_ and_ Decision_ Science/ algorithms_ for_ decision_ making.pdf
- Operations Management/ 08_ TUDelft_ Operations_ Research_ Optimization_ for_ Operations.pdf
- Managerial Accounting/ 13 - Bourne Franco- Santos Micheli Pavlov - Performance Measurement System of Systems.pdf
- Financial Statement Analysis/ 32_ FinAR_ Bench_ Financial_ Statement_ Analysis_ Benchmark_ 2025.pdf

Signal: Requirements, constraints, alternatives, uncertainty, resource limits, performance targets, and observed outcomes.

Meaning: Convert a vague business choice into variables, constraints, assumptions, and an acceptance test.

Value: Fewer impulsive decisions, better model discipline, and faster learning from failed assumptions.

Execution: Write a decision rule that can be tested against actual results.

Practical exercise: Turn a hiring, pricing, or capacity decision into a decision table with assumptions and thresholds.

Mini case study: A founder wants to add a new service line. Model the resource requirements, expected bottlenecks, required margin, and go/no-go trigger.

Deliverable: Decision model v1 with explicit variables and feedback metrics.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define requirements and optimization in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Phase 2: Business Activity into Data

Lesson 5: Transaction Capture to Accounting Memory

Primary folders used: Accounting Information Systems; Financial Accounting; Auditing

Core objective: Build the event-to-record pipeline: source document, journal, ledger, reconciliation, statement.

Key concepts: source document; journal entry; ledger; bank reconciliation; data reliability

Specific files used

- Accounting Information Systems/ 01_ OpenStax_ Principles_ Financial_ Accounting_ AIS_ Chapter_ 7.pdf
- Accounting Information Systems/ 02_ Lumen_ Financial_ Accounting_ AIS_ and_ Bank_ Reconciliation.pdf
- Financial Accounting/ 01 - OpenStax - Principles of Accounting Volume 1 Financial Accounting.pdf
- Financial Accounting/ 22 - SEC - Beginners Guide to Financial Statements.pdf
- Auditing/ GAO_ Assessing_ Data_ Reliability_ 2019.pdf

Signal: Invoices, receipts, bank transactions, contracts, inventory counts, payroll records, and adjusting events.

Meaning: Classify raw events into accounts and test whether the record is complete, accurate, timely, and valid.

Value: Reliable financial memory that supports tax, audit, management, finance, and legal decisions.

Execution: Design a transaction capture workflow with reconciliation and exception handling.

Practical exercise: Take ten sample transactions and trace each to journal entry, ledger impact, statement effect, and audit evidence.

Mini case study: Bank deposits do not match recorded sales. Diagnose whether the problem is timing, missing records, fraud, refunds, or classification error.

Deliverable: Transaction-to-statement trace table.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define source document and journal entry in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 6: Reporting Standards and Machine-Readable Financial Facts

Primary folders used: Financial Accounting; Accounting Information Systems; Auditing

Core objective: Connect recognition, measurement, disclosure, XBRL, and auditability into one reporting system.

Key concepts: recognition; measurement; faithful representation; XBRL; audit evidence

Specific files used

- Financial Accounting/ 12 - FASB - Concepts Statement No 8 As Amended.pdf
- Financial Accounting/ 13 - IFRS - Conceptual Framework for Financial Reporting 2024.pdf
- Accounting Information Systems/ 14_ XBRL_ US_ Financial_ Reporting_ with_ Open_ Data_ Standards.pdf
- Accounting Information Systems/ 15_ XBRL_ International_ Understanding_ XBRL_ for_ Software_ Vendors.pdf
- Auditing/ PCAOB_ Auditing_ Standards_ FY_ Beginning_ After_ 2025- 12- 15.pdf

Signal: Transactions, estimates, disclosures, accounting policies, tagged facts, and audit assertions.

Meaning: Decide which facts qualify for reporting, how they are measured, and whether they are comparable and verifiable.

Value: Decision-useful reports that outside stakeholders can analyze, compare, and trust.

Execution: Build a reporting checklist that links each statement line to source evidence and reporting standard logic.

Practical exercise: Choose one revenue stream and specify recognition rule, measurement basis, disclosure need, XBRL tag logic, and audit assertion.

Mini case study: A company reports strong revenue growth, but contract terms include refunds and performance obligations. Determine what must be recognized now.

Deliverable: Financial reporting assertion matrix.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define recognition and measurement in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 7: Controls, Evidence, and Fraud Signals

Primary folders used: Auditing; Accounting Information Systems; Business Law

Core objective: Treat controls as a feedback system for preventing, detecting, and correcting bad information.

Key concepts: control activity; segregation of duties; fraud risk; audit evidence; continuous monitoring

Specific files used

- Auditing/ GAO_ Green_ Book_ Internal_ Control_ 2025.pdf
- Auditing/ GAO_ Fraud_ Risk_ Management_ Framework_ 2015.pdf
- Auditing/ IAASB_ Handbook_ 2025_ Volume_ 1_ ISAs_ ISQMs.pdf
- Accounting Information Systems/ 10_ MPRA_ Continuous_ Auditing_ Technology_ Review_ 2025.pdf
- Business Law/ 04_ Regulation_ Compliance/ sarbanes- oxley- act- 2002- public- law- 107- 204.pdf

Signal: Exceptions, overrides, unusual journal entries, missing approvals, access logs, failed reconciliations, and complaints.

Meaning: Interpret whether a variance is normal noise, process weakness, intentional manipulation, or legal exposure.

Value: Higher trust, lower loss, better compliance, and earlier detection of system failure.

Execution: Build a control matrix with owner, risk, control, evidence, test, exception rule, and remediation step.

Practical exercise: Design controls over cash receipts, vendor payments, payroll, and marketing spend.

Mini case study: A manager can approve vendors and release payments. Model the fraud pathway and install controls without stopping normal operations.

Deliverable: Internal control and fraud-risk matrix.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define control activity and segregation of duties in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 8: Business Law + AIS: Contracts as Data Systems

Primary folders used: Business Law; Accounting Information Systems; Auditing

Core objective: Model contracts as structured event rules that drive accounting, controls, compliance, and automation.

Key concepts: contract event; obligation; REA model; workflow automation; IT control

Specific files used

- Business Law/ 05_ Scientific_ Empirical_ Law/ contract- as- automaton- computational- representation- financial- agreements.pdf
- Business Law/ 04_ Regulation_ Compliance/ doj- evaluation- of- corporate- compliance- programs- 2024.pdf
- Accounting Information Systems/ 08_ AISel_ REA_ Design_ Theory_ Enterprise_ System_ Design_ 2016.pdf
- Accounting Information Systems/ 18_ RPA_ in_ Accounting_ Applied_ Overview.pdf
- Auditing/ GAO_ FISCAM_ 2024.pdf

Signal: Contract terms, delivery events, payment triggers, exceptions, approvals, and system permissions.

Meaning: Translate legal obligations into data fields, process states, control points, and accounting events.

Value: Lower contract leakage, better revenue capture, cleaner audit trails, and more reliable automation.

Execution: Draft a contract-to-data map for one recurring revenue or vendor agreement.

Practical exercise: Extract obligations, deadlines, amounts, evidence, approvals, and accounting entries from a sample contract.

Mini case study: A subscription contract has renewal, usage, refund, and service-credit clauses. Convert it into system rules that accounting and operations can execute.

Deliverable: Contract event schema and control checklist.

Cross-disciplinary logic: This pairing matters because contracts define the legal event, while AIS determines whether the event becomes reliable data and enforceable execution.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define contract event and obligation in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Phase 3: Data into Financial Meaning

Lesson 9: Ratio Analysis as Business Diagnosis

Primary folders used: Financial Statement Analysis; Financial Accounting; Corporate Finance

Core objective: Convert statements into profitability, liquidity, efficiency, leverage, growth, and valuation meaning.

Key concepts: common-size analysis; trend analysis; liquidity; solvency; valuation comparables

Specific files used

- Financial Statement Analysis/ 01_ Bigel_ Introduction_ to_ Financial_ Analysis_ Open_ Touro_ 2022.pdf
- Financial Statement Analysis/ 04_ MIT_ OCW_ Financial_ Statement_ Analysis_ Ratio_ Analysis_ 2004.pdf
- Financial Statement Analysis/ 06_ NYU_ Ratio_ Analysis_ Common_ Size_ and_ Caveats.pdf
- Financial Accounting/ 14 - Bigel - Introduction to Financial Analysis.pdf
- Corporate Finance/ 18_ Welch_ Valuation_ from_ Comparables_ and_ Ratios_ 2022.pdf

Signal: Income statements, balance sheets, cash flow statements, revenue growth, margins, turnover, debt, and market multiples.

Meaning: Diagnose whether performance changes come from operations, financing, pricing, asset use, or accounting effects.

Value: Better credit, investment, pricing, financing, and management decisions.

Execution: Build a ratio dashboard with comparison base, interpretation rule, and action threshold.

Practical exercise: Use two periods of financial statements to separate margin deterioration from asset-turnover deterioration.

Mini case study: Revenue grows but cash declines and leverage rises. Determine whether the business is scaling profitably or consuming liquidity.

Deliverable: Financial diagnosis dashboard v1.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define common-size analysis and trend analysis in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 10: Quality of Earnings and Distress Detection

Primary folders used: Financial Statement Analysis; Auditing; Financial Accounting

Core objective: Use financial statement patterns to detect manipulation risk, weak fundamentals, and distress.

Key concepts: earnings quality; M-Score; F-Score; Z-Score; fundamental signal

Specific files used

- Financial Statement Analysis/ 12_ Nissim_ Penman_ Ratio_ Analysis_ and_ Equity_ Valuation_ Working_ Paper_ 1999.pdf
- Financial Statement Analysis/ 14_ Nissim_ Penman_ FSA_ of_ Leverage_ Profitability_ Price_ to_ Book_ 2003.pdf
- Financial Statement Analysis/ 16_ Beneish_ Detection_ of_ Earnings_ Manipulation_ 1999.pdf
- Financial Statement Analysis/ 17_ Piotroski_ Value_ Investing_ Historical_ Financial_ Statement_ Information_ 2000.pdf
- Financial Statement Analysis/ 18_ Altman_ Financial_ Ratios_ Discriminant_ Analysis_ Bankruptcy_ 1968.pdf

Signal: Accruals, receivables, margins, leverage, asset quality, cash flow, profitability, and sales growth.

Meaning: Determine whether reported performance is durable, manipulated, distressed, or improving.

Value: Avoid bad investments, bad credit decisions, weak acquisitions, and misleading management reports.

Execution: Create a red/yellow/green diagnostic rule for earnings quality and distress.

Practical exercise: Build a small scoring model using five ratios and write the interpretation rule for each score band.

Mini case study: A target company shows rising sales and profit, but receivables and accruals rise faster. Decide whether due diligence should continue.

Deliverable: Quality-of-earnings risk screen.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define earnings quality and M-Score in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 11: Cost Accounting + Operations: Unit Economics of Flow

Primary folders used: Cost Accounting; Operations Management; Managerial Accounting

Core objective: Connect time, capacity, material flow, bottlenecks, and cost drivers into usable unit economics.

Key concepts: TDABC; material flow cost accounting; cost driver; bottleneck; unit economics

Specific files used

- Cost Accounting/ 13_ Springer_ TDABC_ Costing_ Implementation_ Strategies.pdf
- Cost Accounting/ 09_ APO_ Manual_ on_ Material_ Flow_ Cost_ Accounting_ ISO14051.pdf
- Operations Management/ 10_ Littles_ Law_ 50th_ Anniversary_ Flow_ Science.pdf
- Operations Management/ 13_ Chalmers_ AI_ for_ Throughput_ Bottleneck_ Analysis.pdf
- Managerial Accounting/ 11 - Kaplan Anderson - Time- Driven Activity- Based Costing.pdf

Signal: Minutes, labor rates, machine time, material input, scrap, rework, queues, throughput, and capacity use.

Meaning: Explain which activities consume resources and which constraints make profitable demand unprofitable.

Value: Better pricing, better process improvement, lower waste, and more accurate margin decisions.

Execution: Build a time-driven cost model and use it to rank improvement actions.

Practical exercise: Calculate cost per order before and after a bottleneck improvement; separate real savings from accounting allocation changes.

Mini case study: A product with high gross margin consumes scarce bottleneck time. Decide whether to raise price, reduce demand, redesign process, or drop the product.

Deliverable: Activity cost and bottleneck margin model.

Cross-disciplinary logic: Cost accounting tells what resources are consumed; operations management explains why the resource consumption happens.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define TDABC and material flow cost accounting in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 12: Financial Statement Analysis + System Science: Ratios as Behavior

Primary folders used: Financial Statement Analysis; System Science; Corporate Finance

Core objective: Interpret ratios as symptoms of system behavior, not isolated numbers.

Key concepts: DuPont; behavior over time; process mining; cash flow conversion; ratio validity

Specific files used

- Financial Statement Analysis/ 21_ Linares_ Mustaros_ et_ al_ New_ Financial_ Ratios_ CoDa_ Methodology_ 2022.pdf
- Financial Statement Analysis/ 22_ Saus_ Sala_ et_ al_ Compositional_ DuPont_ Analysis_ 2021.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ behavior.pdf
- System Science/ 04_ Business_ Optimization_ and_ Decision_ Science/ process_ mining_ manifesto.pdf
- Corporate Finance/ 17_ Welch_ From_ Financial_ Statements_ to_ Economic_ Cash_ Flows_ 2022.pdf

Signal: Ratio trends, process events, margin shifts, turnover changes, leverage changes, and cash conversion gaps.

Meaning: Link financial behavior to operating structure, feedback delay, constraint movement, and accounting composition.

Value: More accurate diagnosis and fewer false fixes based on single-ratio thinking.

Execution: Create a causal loop diagram for one deteriorating ratio.

Practical exercise: Trace a falling return on assets ratio back to price, volume, cost, asset turnover, and working capital signals.

Mini case study: A company improves inventory turnover but customer complaints rise. Determine whether the ratio improvement damaged the system.

Deliverable: Ratio-to-causal-loop analysis.

Cross-disciplinary logic: Financial statement analysis identifies the symptom; system science prevents mistaking the symptom for the cause.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define DuPont and behavior over time in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 13: Managerial Accounting + Marketing: Funnel Economics

Primary folders used: Managerial Accounting; Marketing; Accounting Information Systems

Core objective: Connect customer acquisition, retention, CLV, budgets, forecasts, and internal performance control.

Key concepts: CLV; RFM; balanced scorecard; rolling forecast; customer equity

Specific files used

- Managerial Accounting/ 10 - Kaplan - Conceptual Foundations of the Balanced Scorecard.pdf
- Managerial Accounting/ 12 - Henttu- Aho - Rolling Forecasting in Budgetary Control Systems.pdf
- Marketing/ 02_ Market_ Understanding_ Data_ and_ Analytics/ rfm_ customer_ segmentation.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ clv_ insights_ strategic_ marketing.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ customer_ value_ clv_ customer_ equity.pdf

Signal: Leads, conversion rates, CAC, retention, churn, gross margin, purchase frequency, channel spend, and forecast variance.

Meaning: Interpret whether marketing spend creates profitable customer assets or only short-term activity.

Value: Better budget allocation, stronger retention economics, and cleaner growth decisions.

Execution: Build a funnel model that links marketing metrics to contribution margin and forecast updates.

Practical exercise: Calculate customer payback period and decide which segment deserves budget expansion.

Mini case study: Paid ads produce high sales but low repeat purchases. Decide whether to scale, pause, retarget, or redesign the offer.

Deliverable: Marketing economics dashboard.

Cross-disciplinary logic: Marketing creates demand signals; managerial accounting tests whether those signals convert into profitable control targets.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define CLV and RFM in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Phase 4: Financial Meaning into Capital Decisions

Lesson 14: NPV, Hurdle Rates, and Capital Allocation

Primary folders used: Corporate Finance; Financial Statement Analysis; Managerial Accounting

Core objective: Turn forecasted financial meaning into investment rules.

Key concepts: NPV; hurdle rate; forecast; cost of capital; capital rationing

Specific files used

- Corporate Finance/ 09_ Welch_ Present_ Value_ NPV_ 2022.pdf
- Corporate Finance/ 11_ Welch_ Capital_ Budgeting_ Rules_ 2022.pdf
- Corporate Finance/ 14_ Welch_ Benchmarked_ Costs_ of_ Capital_ 2022.pdf
- Financial Statement Analysis/ 20_ Penman_ What_ Matters_ in_ Company_ Valuation_ 2003.pdf
- Managerial Accounting/ 07 - IFAC - Evaluating and Improving Costing in Organizations.pdf

Signal: Projected cash flows, investment cost, risk, capacity, tax effects, margin assumptions, and alternatives.

Meaning: Evaluate whether the project creates value after risk, timing, and opportunity cost.

Value: Capital goes to experiments, assets, and projects with the highest expected value per constrained dollar.

Execution: Build an NPV model with base, downside, and upside cases plus an explicit kill rule.

Practical exercise: Compare two projects with different timing and risk; choose using NPV, payback, and strategic fit.

Mini case study: A company can buy equipment to reduce labor cost or spend the same capital on demand generation. Decide which creates more value.

Deliverable: Capital allocation memo.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define NPV and hurdle rate in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 15: Taxation + Corporate Finance: Tax-Aware Capital Structure

Primary folders used: Taxation; Corporate Finance; Financial Accounting

Core objective: Integrate tax shields, depreciation timing, debt capacity, and financial flexibility.

Key concepts: tax shield; capital structure; depreciation; corporate tax; financial flexibility

Specific files used

- Corporate Finance/ 20_ Welch_ Taxes_ and_ Capital_ Structure_ 2022.pdf
- Corporate Finance/ 28_ Myers_ Capital_ Structure_ Puzzle_ 1984.pdf
- Taxation/ IRS_ Pub_ 542_ Corporations.pdf
- Taxation/ IRS_ Pub_ 946_ How_ to_ Depreciate_ Property.pdf
- Taxation/ JCT_ Overview_ of_ Federal_ Tax_ System_ 2025.pdf

Signal: Debt interest, taxable income, depreciation schedule, projected cash flows, asset collateral, and distress risk.

Meaning: Interpret whether tax benefits justify added financial risk and reduced flexibility.

Value: Higher after-tax value without creating fragility.

Execution: Calculate after-tax project cash flow under alternative debt/equity structures.

Practical exercise: Compare an equipment purchase financed with debt versus equity; include depreciation and interest tax effects.

Mini case study: A business can borrow to buy equipment that improves margin. Test whether tax shields compensate for default risk and demand uncertainty.

Deliverable: Tax-aware financing decision table.

Cross-disciplinary logic: Corporate finance decides how capital is sourced and used; taxation changes the after-tax cash flows that define value.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define tax shield and capital structure in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 16: Entity Choice, Governance, and Capital Access

Primary folders used: Taxation; Business Law; Corporate Finance

Core objective: Choose business structure as an operating, tax, financing, and control decision.

Key concepts: LLC; S corporation; C corporation; agency; investor rights

Specific files used

- Taxation/ IRS_ Pub_ 3402_ Taxation_ of_ LLCs.pdf
- Taxation/ IRS_ Form_ 8832_ Entity_ Classification_ Election.pdf
- Business Law/ 04_ Regulation_ Compliance/ irs- publication- 5868- business- structures.pdf
- Business Law/ 03_ Governance_ Legal_ Structure/ essential- role- of- organizational- law- hansmann- kraakman.pdf
- Corporate Finance/ 32_ Jensen_ Meckling_ Theory_ of_ the_ Firm_ Agency_ Costs_ 1976.pdf

Signal: Ownership goals, profit distribution needs, outside capital needs, tax elections, liability risk, and governance requirements.

Meaning: Interpret how entity choice changes cash flows, control rights, tax compliance, and investor compatibility.

Value: A structure that fits growth strategy instead of creating future friction.

Execution: Design an entity-choice matrix for a business at launch, growth, and outside-investment stages.

Practical exercise: Model how the same $100,000 profit flows through three entity forms.

Mini case study: A creative-services company may raise investor money in 18 months. Decide whether current tax savings are worth possible future restructuring.

Deliverable: Entity and governance roadmap.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define LLC and S corporation in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 17: Corporate Finance + Music Business: Catalog Investment

Primary folders used: Corporate Finance; Music Business; Financial Statement Analysis

Core objective: Value a music rights asset as a risky cash-flow stream.

Key concepts: royalty cash flow; catalog valuation; discount rate; scenario analysis; asset risk

Specific files used

- Music Business/ 02_ Revenue_ and_ Streaming/ 01_ WIPO_ IP_ Finance_ in_ the_ Music_ Industry_ 2026.pdf
- Music Business/ 02_ Revenue_ and_ Streaming/ 02_ WIPO_ Measuring_ IP_ Finance_ and_ Investment_ in_ Music_ 2026.pdf
- Corporate Finance/ 05_ Damodaran_ Valuation_ Approaches_ and_ Metrics_ 2006.pdf
- Corporate Finance/ 09_ Welch_ Present_ Value_ NPV_ 2022.pdf
- Financial Statement Analysis/ 20_ Penman_ What_ Matters_ in_ Company_ Valuation_ 2003.pdf

Signal: Streams, royalties, ownership splits, term, territory, platform dependence, marketing lift, and historical cash flow.

Meaning: Interpret durability, concentration risk, rights quality, and forecast uncertainty.

Value: Avoid overpaying for creative assets and identify which rights can compound value.

Execution: Build a rights-investment NPV model with sensitivity to stream growth, royalty rate, and discount rate.

Practical exercise: Value a catalog using three revenue scenarios and a minimum required return.

Mini case study: An investor can buy a 50% publishing interest in a small catalog. Decide the maximum price and what diligence evidence is required.

Deliverable: Music catalog investment memo.

Cross-disciplinary logic: Music assets produce irregular rights-based cash flows; corporate finance supplies the valuation and risk discipline.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define royalty cash flow and catalog valuation in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 18: Taxation + Music Business: Royalty Cash Systems

Primary folders used: Taxation; Music Business; Business Law

Core objective: Map music revenue rights into taxable income, deductible costs, contracts, and cash timing.

Key concepts: royalties; copyright; deductible expense; contract term; cash timing

Specific files used

- Music Business/ 01_ IP_ and_ Rights/ 03_ USCO_ Copyright_ and_ the_ Music_ Marketplace.pdf
- Music Business/ 02_ Revenue_ and_ Streaming/ 05_ MMF_ Song_ Royalties_ Guide.pdf
- Taxation/ IRS_ Pub_ 334_ Tax_ Guide_ for_ Small_ Business.pdf
- Taxation/ IRS_ Pub_ 535_ Business_ Expenses_ Archived_ 2022.pdf
- Business Law/ 02_ Contracts/ sales- and- leases- ucc- article- 2- and- 2a- cali.pdf

Signal: Royalty statements, publishing splits, master rights, advances, recoupment, contracts, expenses, and payment timing.

Meaning: Interpret who earned what, when it is recognized, what costs are deductible, and what evidence supports the position.

Value: Higher after-tax cash retention, cleaner rights accounting, and lower audit risk.

Execution: Create a royalty income and expense classification map for an artist or small label.

Practical exercise: Classify ten music-business cash flows as royalty income, service income, advance, reimbursable cost, asset cost, or deductible expense.

Mini case study: An artist receives an advance, streaming royalties, merch income, and tour support. Build the tax-aware cash map.

Deliverable: Royalty tax and documentation checklist.

Cross-disciplinary logic: Music-business value often leaks through rights complexity and tax timing; taxation converts those rights into real cash consequences.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define royalties and copyright in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Phase 5: Market Demand and Revenue Systems

Lesson 19: Market Demand as a Measurement System

Primary folders used: Marketing; Managerial Accounting; System Science

Core objective: Treat marketing as the sensing system for customer demand and willingness to pay.

Key concepts: segmentation; buyer behavior; value proposition; marketing analytics; customer signal

Specific files used

- Marketing/ INDEX.md
- Marketing/ 01_ Foundations_ Purpose_ and_ Market_ Logic/ openstax_ principles_ of_ marketing.pdf
- Marketing/ 01_ Foundations_ Purpose_ and_ Market_ Logic/ intro_ consumer_ behaviour.pdf
- Marketing/ 02_ Market_ Understanding_ Data_ and_ Analytics/ marketing_ analytics_ methods_ practice.pdf
- Managerial Accounting/ 09 - CGMA - Global Management Accounting Principles.pdf

Signal: Search behavior, social response, sales conversations, conversion, churn, price response, and customer complaints.

Meaning: Interpret which customer problems are urgent, measurable, reachable, and profitable.

Value: Better offers, stronger positioning, less wasted spend, and higher revenue quality.

Execution: Design a demand test with a measurable success threshold.

Practical exercise: Create three customer hypotheses and define the signal that would validate or reject each one.

Mini case study: A product gets many likes but few purchases. Determine whether the issue is audience, offer, channel, price, or trust.

Deliverable: Demand signal test plan.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define segmentation and buyer behavior in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 20: Music Business + System Science: Audience Feedback Loops

Primary folders used: Music Business; System Science; Marketing

Core objective: Model fan data, recommendation systems, social response, and royalty outcomes as feedback loops.

Key concepts: fan data; network effects; social demand; platform feedback; leverage point

Specific files used

- Music Business/ MANIFEST.md
- Music Business/ 03_ Audience_ and_ Fan_ Data/ 01_ MMF_ Fan_ Data_ Guide_ 2026.pdf
- Music Business/ 03_ Audience_ and_ Fan_ Data/ 05_ Arxiv_ Impact_ of_ Social_ Media_ on_ Music_ Demand.pdf
- System Science/ 01_ Foundations_ and_ Complex_ Systems/ networks_ crowds_ markets.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ meadows_ leverage_ points.pdf

Signal: Streams, saves, follows, shares, playlist adds, merch purchases, email signups, geography, and repeat engagement.

Meaning: Interpret whether demand is broadening, deepening, or decaying through platform and audience feedback.

Value: Better release strategy, touring choices, fan ownership, and revenue compounding.

Execution: Build an audience feedback loop map with one owned-data leverage point.

Practical exercise: Map how one release converts content signals into audience assets and future revenue.

Mini case study: A track gets algorithmic streams but weak fan conversion. Decide how to shift from platform exposure to owned audience.

Deliverable: Music audience system map.

Cross-disciplinary logic: Music demand is networked and feedback-driven; system science shows why isolated campaign metrics can mislead.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define fan data and network effects in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 21: Auditing + Marketing: Evidence for Growth Claims

Primary folders used: Auditing; Marketing; Business Law

Core objective: Audit the evidence behind marketing performance, attribution, claims, and compliance.

Key concepts: attribution; data reliability; continuous monitoring; claim substantiation; marketing compliance

Specific files used

- Marketing/ 02_ Market_ Understanding_ Data_ and_ Analytics/ marketing_ mix_ modeling_ bayesian.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ legal_ aspects_ marketing_ sales.pdf
- Auditing/ GAO_ Assessing_ Data_ Reliability_ 2019.pdf
- Auditing/ NIST_ SP_ 800- 137_ Continuous_ Monitoring.pdf
- Business Law/ 04_ Regulation_ Compliance/ ftc- doj- antitrust- guidelines- business- activities- affecting- workers- 2025.pdf

Signal: Ad spend, impressions, clicks, conversions, lift estimates, offer claims, customer data, and channel reports.

Meaning: Interpret which growth signals are causal, which are vanity metrics, and which create legal or control risk.

Value: More reliable growth decisions and lower risk of waste, misstatement, or illegal claims.

Execution: Create an audit checklist for marketing data and promotional claims.

Practical exercise: Test whether a campaign's reported ROI is supported by source data, attribution logic, and customer outcome evidence.

Mini case study: Marketing reports 4x ROAS, but sales team says most conversions were existing customers. Audit the claim before increasing spend.

Deliverable: Marketing evidence audit checklist.

Cross-disciplinary logic: Marketing creates persuasive claims; auditing tests whether the claims are evidence-based and decision-safe.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define attribution and data reliability in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 22: Revenue Operations: From Funnel to Fulfillment

Primary folders used: Marketing; Operations Management; Accounting Information Systems; Financial Accounting

Core objective: Connect acquisition, sales, delivery, invoicing, revenue recognition, and dashboard control.

Key concepts: revenue operations; sales pipeline; fulfillment; dashboard; revenue recognition

Specific files used

- Marketing/ 03_ Value_ Brand_ and_ Communication_ Systems/ emarketing_ 7e.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ power_ of_ selling.pdf
- Operations Management/ 21_ EPA_ Lean_ and_ Clean_ Value_ Stream_ Mapping.pdf
- Accounting Information Systems/ 13_ AsianInstitute_ Data_ Dashboarding_ in_ Accounting_ Tableau_ 2023.pdf
- Financial Accounting/ 22 - SEC - Beginners Guide to Financial Statements.pdf

Signal: Lead source, deal stage, close rate, delivery status, invoice date, cash receipt, refunds, and support load.

Meaning: Interpret whether revenue growth is operationally fulfillable, collectible, and reportable.

Value: Growth that does not break delivery, cash flow, or accounting integrity.

Execution: Build a revenue operations flow from first touch to cash and post-sale feedback.

Practical exercise: Trace five customer orders through marketing, sales, operations, accounting, and service feedback.

Mini case study: A campaign doubles leads but delivery time increases and refunds rise. Decide whether to scale demand or stabilize operations.

Deliverable: Revenue operations control map.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define revenue operations and sales pipeline in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Phase 6: Integrated Business Science Labs

Lesson 23: Integrated Profit Diagnosis

Primary folders used: Financial Accounting; Managerial Accounting; Cost Accounting; Operations Management; Financial Statement Analysis

Core objective: Diagnose low profit by separating price, volume, mix, cost, capacity, waste, and accounting classification.

Key concepts: contribution margin; variance; cost driver; bottleneck; ratio diagnosis

Specific files used

- Financial Accounting/ 01 - OpenStax - Principles of Accounting Volume 1 Financial Accounting.pdf
- Managerial Accounting/ 02 - Heisinger Hoyle - Managerial Accounting.pdf
- Cost Accounting/ 21_ IFAC_ Evaluating_ and_ Improving_ Costing_ in_ Organizations.pdf
- Financial Statement Analysis/ 08_ AICPA_ Analyzing_ Financial_ Ratios_ Practice_ Aid_ 2006.pdf
- Operations Management/ 20_ JIEM_ Theory_ of_ Constraints_ Make_ to_ Order_ Case_ Study.pdf

Signal: Sales by product, direct costs, overhead, capacity use, WIP, rework, gross margin, operating margin, and cash conversion.

Meaning: Identify whether low profit is caused by market, operations, cost model, financial structure, or reporting distortion.

Value: A prioritized improvement plan instead of random cost cutting.

Execution: Build a profit bridge and choose the highest-leverage intervention.

Practical exercise: Create a waterfall from revenue to operating profit and annotate each variance with a likely cause.

Mini case study: A business sells more units but profit declines. Diagnose price erosion, unfavorable mix, bottleneck overtime, and overhead absorption effects.

Deliverable: Profit diagnosis report.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define contribution margin and variance in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 24: Auditing + Operations: Control the Workflow

Primary folders used: Auditing; Operations Management; Accounting Information Systems

Core objective: Use audit logic to test whether an operating workflow is controlled, measured, and improving.

Key concepts: control test; digital twin; continuous monitoring; exception rule; corrective action

Specific files used

- Auditing/ GAO_ Green_ Book_ Internal_ Control_ 2025.pdf
- Auditing/ NIST_ SP_ 800- 53Ar5_ Assessing_ Security_ Privacy_ Controls.pdf
- Operations Management/ 14_ DiVA_ Digital_ Twin_ Bottleneck_ Diagnosis_ and_ Throughput_ Improvement.pdf
- Operations Management/ 24_ MDPI_ I3oT_ Sub_ Bottleneck_ Detection_ Continuous_ Improvement.pdf
- Accounting Information Systems/ 10_ MPRA_ Continuous_ Auditing_ Technology_ Review_ 2025.pdf

Signal: Process logs, access logs, control evidence, exception rates, bottleneck data, and corrective-action history.

Meaning: Interpret whether workflow problems are random variation, weak controls, poor incentives, or bad system design.

Value: More reliable operations and fewer hidden failures.

Execution: Design an operational control test with evidence, frequency, owner, and remediation trigger.

Practical exercise: Audit a purchase-to-pay or order-to-cash workflow for three failure modes.

Mini case study: A production cell reports good averages but late orders cluster around one shift. Test whether this is capacity, data, staffing, or control failure.

Deliverable: Operational audit program.

Cross-disciplinary logic: Operations produces the work; auditing tests whether the work system is trustworthy and controlled.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define control test and digital twin in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 25: Cash Flow vs Profit

Primary folders used: Financial Accounting; Corporate Finance; Taxation; Financial Statement Analysis

Core objective: Explain why profit, taxable income, and cash flow diverge, then make a decision from the cash reality.

Key concepts: accrual accounting; working capital; free cash flow; tax timing; earnings quality

Specific files used

- Financial Accounting/ 22 - SEC - Beginners Guide to Financial Statements.pdf
- Corporate Finance/ 17_ Welch_ From_ Financial_ Statements_ to_ Economic_ Cash_ Flows_ 2022.pdf
- Corporate Finance/ 23_ Welch_ Pro_ Forma_ Financial_ Statements_ 2022.pdf
- Taxation/ IRS_ Pub_ 538_ Accounting_ Periods_ and_ Methods.pdf
- Financial Statement Analysis/ 02_ Penman_ The_ Quality_ of_ Financial_ Statements_ 2003.pdf

Signal: Revenue, receivables, payables, inventory, depreciation, taxable income, estimated payments, and operating cash flow.

Meaning: Interpret timing differences and decide whether profit is converting into cash.

Value: Avoid growth that looks profitable but creates liquidity stress.

Execution: Build a cash conversion bridge from net income to operating cash flow to free cash flow.

Practical exercise: Reconcile accounting profit to cash and identify the top three cash drains.

Mini case study: A business shows positive net income but cannot meet payroll. Diagnose working capital, tax timing, capital spending, and collection failures.

Deliverable: Cash conversion dashboard.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define accrual accounting and working capital in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Lesson 26: The Business Science Operating System

Primary folders used: Accounting Information Systems; Auditing; Business Law; Corporate Finance; Cost Accounting; Financial Accounting; Financial Statement Analysis; Managerial Accounting; Marketing; Music Business; Operations Management; System Science; Taxation

Core objective: Assemble the full business science loop: observe, interpret, decide, execute, control, and improve.

Key concepts: operating system; dashboard; control loop; capstone; continuous improvement

Specific files used

- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ checklist.pdf
- Accounting Information Systems/ 20_ WorldBank_ Financial_ Management_ Information_ System_ AIS_ ERP_ Dashboards.pdf
- Managerial Accounting/ 13 - Bourne Franco- Santos Micheli Pavlov - Performance Measurement System of Systems.pdf
- Business Law/ 04_ Regulation_ Compliance/ doj- evaluation- of- corporate- compliance- programs- 2024.pdf
- Music Business/ 04_ Market_ and_ Ecosystem_ Reports/ 01_ IFPI_ Global_ Music_ Report_ 2026_ State_ of_ the_ Industry.pdf

Signal: All business signals: market, operations, accounting, cost, finance, tax, audit, law, and music-industry data.

Meaning: Interpret how each subsystem changes the others through constraints, incentives, cash, risk, and feedback.

Value: A complete management system for making money through evidence and disciplined improvement.

Execution: Build the capstone architecture and run one improvement cycle.

Practical exercise: Create a system map with one metric per folder and one decision rule per metric.

Mini case study: A small music venture sells services, owns rights, runs campaigns, pays contractors, and considers a catalog investment. Build its operating system.

Deliverable: Business Science Operating System blueprint.

Questions

- Recall: What raw signals does this lesson start from, and which source files define those signals?
- Recall: Define operating system and dashboard in operational business terms.
- Application: Take one real or hypothetical transaction/event and map it through signal -> meaning -> value -> execution.
- Application: What metric would you calculate first, and what threshold would change your action?
- Systems-thinking: Where is the feedback loop, delay, or constraint that could make the obvious fix fail?
- Business decision: If the evidence is mixed, what decision would you make now, defer, or test next?
- Reflection: What assumption in your interpretation is most fragile, and how would you test it cheaply?

Cross-Disciplinary Lesson Map

Lesson	Pairing	Folders	Why it matters
L8	Business Law + AIS: Contracts as Data Systems	Business Law; Accounting Information Systems; Auditing	This pairing matters because contracts define the legal event, while AIS determines whether the event becomes reliable data and enforceable execution.
L11	Cost Accounting + Operations: Unit Economics of Flow	Cost Accounting; Operations Management; Managerial Accounting	Cost accounting tells what resources are consumed; operations management explains why the resource consumption happens.
L12	Financial Statement Analysis + System Science: Ratios as Behavior	Financial Statement Analysis; System Science; Corporate Finance	Financial statement analysis identifies the symptom; system science prevents mistaking the symptom for the cause.
L13	Managerial Accounting + Marketing: Funnel Economics	Managerial Accounting; Marketing; Accounting Information Systems	Marketing creates demand signals; managerial accounting tests whether those signals convert into profitable control targets.
L15	Taxation + Corporate Finance: Tax-Aware Capital Structure	Taxation; Corporate Finance; Financial Accounting	Corporate finance decides how capital is sourced and used; taxation changes the after-tax cash flows that define value.
L17	Corporate Finance + Music Business: Catalog Investment	Corporate Finance; Music Business; Financial Statement Analysis	Music assets produce irregular rights-based cash flows; corporate finance supplies the valuation and risk discipline.
L18	Taxation + Music Business: Royalty Cash Systems	Taxation; Music Business; Business Law	Music-business value often leaks through rights complexity and tax timing; taxation converts those rights into real cash consequences.
L20	Music Business + System Science: Audience Feedback Loops	Music Business; System Science; Marketing	Music demand is networked and feedback-driven; system science shows why isolated campaign metrics can mislead.
L21	Auditing + Marketing: Evidence for Growth Claims	Auditing; Marketing; Business Law	Marketing creates persuasive claims; auditing tests whether the claims are evidence-based and decision-safe.
L24	Auditing + Operations: Control the Workflow	Auditing; Operations Management; Accounting Information Systems	Operations produces the work; auditing tests whether the work system is trustworthy and controlled.

Business Science Labs

Lab 1: Build a Simple Business Dashboard

Business problem: Management cannot see demand, fulfillment, cash, margin, and risk in one place.

Source folders used: Accounting Information Systems; Managerial Accounting; Marketing; Operations Management; Financial Accounting

Data or documents needed: Sales log, customer pipeline, fulfillment status, bank activity, budget, and chart of accounts.

Analysis process: Define metric owner -> map source field -> test data quality -> calculate KPI -> set threshold -> review weekly.

Decision rule: A metric belongs on the dashboard only if it changes a decision within one review cycle.

Final deliverable: One-page dashboard specification with signal, owner, formula, threshold, and action.

Lab 2: Diagnose Why Profit Is Low

Business problem: Revenue is growing, but operating profit and cash are weak.

Source folders used: Financial Statement Analysis; Cost Accounting; Managerial Accounting; Operations Management; Corporate Finance

Data or documents needed: Income statement, product mix, cost drivers, capacity data, price history, and cash flow statement.

Analysis process: Build profit bridge -> isolate mix/price/cost/capacity effects -> test bottleneck -> identify highest leverage fix.

Decision rule: Act first on the cause with the largest recurring margin impact that is controllable within 30 days.

Final deliverable: Profit diagnosis memo with ranked interventions.

Lab 3: Analyze a Music Business Investment

Business problem: Decide whether a catalog, artist advance, or marketing campaign is worth funding.

Source folders used: Music Business; Corporate Finance; Financial Statement Analysis; Taxation; Business Law

Data or documents needed: Royalty history, rights split, contract terms, expected marketing lift, tax treatment, and discount rate.

Analysis process: Validate rights -> forecast cash flows -> apply tax and recoupment -> discount by risk -> stress-test concentration.

Decision rule: Invest only if downside NPV is not catastrophic and base-case IRR beats the hurdle rate after tax.

Final deliverable: Music investment memo with maximum price and diligence checklist.

Lab 4: Create a Tax-Aware Pricing Strategy

Business problem: Pricing decisions ignore sales mix, deductible costs, entity tax effects, and cash timing.

Source folders used: Taxation; Cost Accounting; Managerial Accounting; Marketing; Corporate Finance

Data or documents needed: Price list, unit cost, channel fees, expected volume, deductible expenses, and tax rate assumptions.

Analysis process: Calculate contribution margin -> model taxable income -> compare after-tax cash by price tier -> run demand sensitivity.

Decision rule: Choose the price that maximizes expected after-tax contribution subject to demand and capacity constraints.

Final deliverable: Tax-aware price and margin model.

Lab 5: Build an Audit Checklist for a Small Business

Business problem: The business lacks evidence discipline around cash, payroll, vendors, taxes, and marketing spend.

Source folders used: Auditing; Accounting Information Systems; Financial Accounting; Taxation; Business Law

Data or documents needed: Bank records, invoices, payroll reports, tax filings, contracts, approvals, and access logs.

Analysis process: Map risk -> define assertion -> identify evidence -> test control -> record exception -> assign remediation.

Decision rule: Every material cash movement needs an owner, source document, approval path, and reconciliation.

Final deliverable: Audit checklist with test steps and exception escalation.

Lab 6: Map an Operations Bottleneck

Business problem: Lead times are increasing and managers disagree about the constraint.

Source folders used: Operations Management; System Science; Cost Accounting; Managerial Accounting; Auditing

Data or documents needed: WIP, cycle times, queue times, defect rates, staffing, machine time, and cost per activity.

Analysis process: Map flow -> apply Little's Law -> find queue accumulation -> quantify constraint cost -> test with evidence.

Decision rule: A bottleneck is real only if relieving it increases system throughput or reduces delay without shifting failure elsewhere.

Final deliverable: Bottleneck map with measured constraint and improvement experiment.

Lab 7: Compare Cash Flow vs Profit

Business problem: Financial statements show profit, but available cash keeps shrinking.

Source folders used: Financial Accounting; Financial Statement Analysis; Corporate Finance; Taxation; Accounting Information Systems

Data or documents needed: Income statement, balance sheet, cash flow statement, receivables aging, payables aging, and tax payment schedule.

Analysis process: Reconcile net income to operating cash -> isolate working capital -> include tax timing -> calculate free cash flow.

Decision rule: Growth is healthy only when cash conversion improves or the financing plan explicitly funds the gap.

Final deliverable: Cash conversion bridge and liquidity action plan.

Lab 8: Design an Accounting Information System

Business problem: Data is scattered across bank feeds, spreadsheets, invoices, contracts, and dashboards.

Source folders used: Accounting Information Systems; Financial Accounting; Auditing; Business Law; Operations Management

Data or documents needed: Chart of accounts, transaction types, contract events, approval roles, source documents, and reporting needs.

Analysis process: Define event schema -> map accounts -> set controls -> reconcile external evidence -> build dashboard outputs.

Decision rule: No data field enters the system unless it supports classification, control, decision, or compliance.

Final deliverable: AIS blueprint with event fields, controls, reports, and feedback loops.

Lab 9: Create a Marketing Funnel with Financial Controls

Business problem: Campaign decisions are made from clicks and sales, not contribution, retention, cash, and evidence.

Source folders used: Marketing; Managerial Accounting; Financial Statement Analysis; Auditing; Accounting Information Systems

Data or documents needed: Ad spend, impressions, leads, conversion, CAC, gross margin, repeat purchase, refund rate, and source evidence.

Analysis process: Map funnel -> calculate unit economics -> audit attribution -> forecast retention -> set scale/pause thresholds.

Decision rule: Scale a channel only when contribution margin, payback period, and evidence quality all pass thresholds.

Final deliverable: Controlled funnel economics model.

Lab 10: Build a System Improvement Plan

Business problem: The business fixes symptoms but does not learn from repeated failure patterns.

Source folders used: Accounting Information Systems; Auditing; Business Law; Corporate Finance; Cost Accounting; Financial Accounting; Financial Statement Analysis; Managerial Accounting; Marketing; Music Business; Operations Management; System Science; Taxation

Data or documents needed: Dashboard metrics, financial statements, cost model, tax calendar, audit exceptions, legal risks, marketing results, and music-business revenue data.

Analysis process: Define recurring signal -> diagnose root cause -> choose leverage point -> run experiment -> measure value -> update system.

Decision rule: An improvement is accepted only if it changes a metric, reduces risk, or increases after-tax cash without damaging another subsystem.

Final deliverable: 90-day system improvement plan with experiments, owners, thresholds, and review cadence.

Coverage Matrix

Subfolder	Lessons using it	Labs using it	Used enough
Accounting Information Systems	L5, L6, L7, L8, L13, L22, L24, L26	Lab 1, Lab 5, Lab 7, Lab 8, Lab 9, Lab 10	Yes
Auditing	L5, L6, L7, L8, L10, L21, L24, L26	Lab 5, Lab 6, Lab 8, Lab 9, Lab 10	Yes
Business Law	L1, L2, L3, L7, L8, L16, L18, L21, L26	Lab 3, Lab 5, Lab 8, Lab 10	Yes
Corporate Finance	L3, L9, L12, L14, L15, L16, L17, L25, L26	Lab 2, Lab 3, Lab 4, Lab 7, Lab 10	Yes
Cost Accounting	L11, L23, L26	Lab 2, Lab 4, Lab 6, Lab 10	Yes
Financial Accounting	L5, L6, L9, L10, L15, L22, L23, L25, L26	Lab 1, Lab 5, Lab 7, Lab 8, Lab 10	Yes
Financial Statement Analysis	L9, L10, L12, L14, L17, L23, L25, L26	Lab 2, Lab 3, Lab 7, Lab 9, Lab 10	Yes
Managerial Accounting	L4, L11, L13, L14, L19, L23, L26	Lab 1, Lab 2, Lab 4, Lab 6, Lab 9, Lab 10	Yes
Marketing	L13, L19, L20, L21, L22, L26	Lab 1, Lab 4, Lab 9, Lab 10	Yes
Music Business	L17, L18, L20, L26	Lab 3, Lab 10	Yes
Operations Management	L1, L2, L4, L11, L22, L23, L24, L26	Lab 1, Lab 2, Lab 6, Lab 8, Lab 10	Yes
System Science	L1, L2, L4, L12, L19, L20, L26	Lab 6, Lab 10	Yes
Taxation	L3, L15, L16, L18, L25, L26	Lab 3, Lab 4, Lab 5, Lab 7, Lab 10	Yes

Source Coverage by Lesson

The files below are the explicit lesson sources. The full 457-file source index is exported separately as CSV.

L1: Business Science Operating Grammar

- System Science/ INDEX.md
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ meadows_ leverage_ points.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ feedback.pdf
- Operations Management/ Operations_ Management_ PDF_ Index.md
- Business Law/ business_ law_ pdf_ index.md

L2: Operations Activity Engine

- Operations Management/ 01_ Seneca_ Introduction_ to_ Operations_ Management_ Process_ Capacity.pdf
- Operations Management/ 10_ Littles_ Law_ 50th_ Anniversary_ Flow_ Science.pdf
- Operations Management/ 21_ EPA_ Lean_ and_ Clean_ Value_ Stream_ Mapping.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ behavior.pdf
- Business Law/ 05_ Scientific_ Empirical_ Law/ formal- methods- business- process- compliance- systematic- review.pdf

L3: Legal Structure as System Boundary

- Business Law/ 01_ Core_ Textbooks/ business- law- and- the- legal- environment- saylor.pdf
- Business Law/ 03_ Governance_ Legal_ Structure/ essential- role- of- organizational- law- hansmann- kraakman.pdf
- Business Law/ 03_ Governance_ Legal_ Structure/ g20- oecd- principles- of- corporate- governance- 2023.pdf
- Taxation/ IRS_ Pub_ 3402_ Taxation_ of_ LLCs.pdf
- Corporate Finance/ 32_ Jensen_ Meckling_ Theory_ of_ the_ Firm_ Agency_ Costs_ 1976.pdf

L4: Decision Models and System Design

- System Science/ 03_ System_ Design_ and_ Engineering/ nasa_ systems_ engineering_ handbook.pdf
- System Science/ 04_ Business_ Optimization_ and_ Decision_ Science/ algorithms_ for_ decision_ making.pdf
- Operations Management/ 08_ TUDelft_ Operations_ Research_ Optimization_ for_ Operations.pdf
- Managerial Accounting/ 13 - Bourne Franco- Santos Micheli Pavlov - Performance Measurement System of Systems.pdf
- Financial Statement Analysis/ 32_ FinAR_ Bench_ Financial_ Statement_ Analysis_ Benchmark_ 2025.pdf

L5: Transaction Capture to Accounting Memory

- Accounting Information Systems/ 01_ OpenStax_ Principles_ Financial_ Accounting_ AIS_ Chapter_ 7.pdf
- Accounting Information Systems/ 02_ Lumen_ Financial_ Accounting_ AIS_ and_ Bank_ Reconciliation.pdf
- Financial Accounting/ 01 - OpenStax - Principles of Accounting Volume 1 Financial Accounting.pdf
- Financial Accounting/ 22 - SEC - Beginners Guide to Financial Statements.pdf

- Auditing/ GAO_ Assessing_ Data_ Reliability_ 2019.pdf

L6: Reporting Standards and Machine-Readable Financial Facts

- Financial Accounting/ 12 - FASB - Concepts Statement No 8 As Amended.pdf
- Financial Accounting/ 13 - IFRS - Conceptual Framework for Financial Reporting 2024.pdf
- Accounting Information Systems/ 14_ XBRL_ US_ Financial_ Reporting_ with_ Open_ Data_ Standards.pdf
- Accounting Information Systems/ 15_ XBRL_ International_ Understanding_ XBRL_ for_ Software_ Vendors.pdf
- Auditing/ PCAOB_ Auditing_ Standards_ FY_ Beginning_ After_ 2025- 12- 15.pdf

L7: Controls, Evidence, and Fraud Signals

- Auditing/ GAO_ Green_ Book_ Internal_ Control_ 2025.pdf
- Auditing/ GAO_ Fraud_ Risk_ Management_ Framework_ 2015.pdf
- Auditing/ IAASB_ Handbook_ 2025_ Volume_ 1_ ISAs_ ISQMs.pdf
- Accounting Information Systems/ 10_ MPRA_ Continuous_ Auditing_ Technology_ Review_ 2025.pdf
- Business Law/ 04_ Regulation_ Compliance/ sarbanes- oxley- act- 2002- public- law- 107- 204.pdf

L8: Business Law + AIS: Contracts as Data Systems

- Business Law/ 05_ Scientific_ Empirical_ Law/ contract- as- automaton- computational- representation- financial- agreements.pdf
- Business Law/ 04_ Regulation_ Compliance/ doj- evaluation- of- corporate- compliance- programs- 2024.pdf
- Accounting Information Systems/ 08_ AISel_ REA_ Design_ Theory_ Enterprise_ System_ Design_ 2016.pdf
- Accounting Information Systems/ 18_ RPA_ in_ Accounting_ Applied_ Overview.pdf
- Auditing/ GAO_ FISCAM_ 2024.pdf

L9: Ratio Analysis as Business Diagnosis

- Financial Statement Analysis/ 01_ Bigel_ Introduction_ to_ Financial_ Analysis_ Open_ Touro_ 2022.pdf
- Financial Statement Analysis/ 04_ MIT_ OCW_ Financial_ Statement_ Analysis_ Ratio_ Analysis_ 2004.pdf
- Financial Statement Analysis/ 06_ NYU_ Ratio_ Analysis_ Common_ Size_ and_ Caveats.pdf
- Financial Accounting/ 14 - Bigel - Introduction to Financial Analysis.pdf
- Corporate Finance/ 18_ Welch_ Valuation_ from_ Comparables_ and_ Ratios_ 2022.pdf

L10: Quality of Earnings and Distress Detection

- Financial Statement Analysis/ 12_ Nissim_ Penman_ Ratio_ Analysis_ and_ Equity_ Valuation_ Working_ Paper_ 1999.pdf
- Financial Statement Analysis/ 14_ Nissim_ Penman_ FSA_ of_ Leverage_ Profitability_ Price_ to_ Book_ 2003.pdf
- Financial Statement Analysis/ 16_ Beneish_ Detection_ of_ Earnings_ Manipulation_ 1999.pdf
- Financial Statement Analysis/ 17_ Piotroski_ Value_ Investing_ Historical_ Financial_ Statement_ Information_ 2000.pdf
- Financial Statement Analysis/ 18_ Altman_ Financial_ Ratios_ Discriminant_ Analysis_ Bankruptcy_ 1968.pdf

L11: Cost Accounting + Operations: Unit Economics of Flow

- Cost Accounting/ 13_ Springer_ TDABC_ Costing_ Implementation_ Strategies.pdf
- Cost Accounting/ 09_ APO_ Manual_ on_ Material_ Flow_ Cost_ Accounting_ ISO14051.pdf
- Operations Management/ 10_ Littles_ Law_ 50th_ Anniversary_ Flow_ Science.pdf
- Operations Management/ 13_ Chalmers_ AI_ for_ Throughput_ Bottleneck_ Analysis.pdf
- Managerial Accounting/ 11 - Kaplan Anderson - Time- Driven Activity- Based Costing.pdf

L12: Financial Statement Analysis + System Science: Ratios as Behavior

- Financial Statement Analysis/ 21_ Linares_ Mustaros_ et_ al_ New_ Financial_ Ratios_ CoDa_ Methodology_ 2022.pdf
- Financial Statement Analysis/ 22_ Saus_ Sala_ et_ al_ Compositional_ DuPont_ Analysis_ 2021.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ behavior.pdf
- System Science/ 04_ Business_ Optimization_ and_ Decision_ Science/ process_ mining_ manifesto.pdf
- Corporate Finance/ 17_ Welch_ From_ Financial_ Statements_ to_ Economic_ Cash_ Flows_ 2022.pdf

L13: Managerial Accounting + Marketing: Funnel Economics

- Managerial Accounting/ 10 - Kaplan - Conceptual Foundations of the Balanced Scorecard.pdf
- Managerial Accounting/ 12 - Henttu- Aho - Rolling Forecasting in Budgetary Control Systems.pdf

- Marketing/ 02_ Market_ Understanding_ Data_ and_ Analytics/ rfm_ customer_ segmentation.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ clv_ insights_ strategic_ marketing.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ customer_ value_ clv_ customer_ equity.pdf

L14: NPV, Hurdle Rates, and Capital Allocation

- Corporate Finance/ 09_ Welch_ Present_ Value_ NPV_ 2022.pdf
- Corporate Finance/ 11_ Welch_ Capital_ Budgeting_ Rules_ 2022.pdf
- Corporate Finance/ 14_ Welch_ Benchmarked_ Costs_ of_ Capital_ 2022.pdf
- Financial Statement Analysis/ 20_ Penman_ What_ Matters_ in_ Company_ Valuation_ 2003.pdf
- Managerial Accounting/ 07 - IFAC - Evaluating and Improving Costing in Organizations.pdf

L15: Taxation + Corporate Finance: Tax-Aware Capital Structure

- Corporate Finance/ 20_ Welch_ Taxes_ and_ Capital_ Structure_ 2022.pdf
- Corporate Finance/ 28_ Myers_ Capital_ Structure_ Puzzle_ 1984.pdf
- Taxation/ IRS_ Pub_ 542_ Corporations.pdf
- Taxation/ IRS_ Pub_ 946_ How_ to_ Depreciate_ Property.pdf
- Taxation/ JCT_ Overview_ of_ Federal_ Tax_ System_ 2025.pdf

L16: Entity Choice, Governance, and Capital Access

- Taxation/ IRS_ Pub_ 3402_ Taxation_ of_ LLCs.pdf
- Taxation/ IRS_ Form_ 8832_ Entity_ Classification_ Election.pdf
- Business Law/ 04_ Regulation_ Compliance/ irs- publication- 5868- business- structures.pdf
- Business Law/ 03_ Governance_ Legal_ Structure/ essential- role- of- organizational- law- hansmann- kraakman.pdf
- Corporate Finance/ 32_ Jensen_ Meckling_ Theory_ of_ the_ Firm_ Agency_ Costs_ 1976.pdf

L17: Corporate Finance + Music Business: Catalog Investment

- Music Business/ 02_ Revenue_ and_ Streaming/ 01_ WIPO_ IP_ Finance_ in_ the_ Music_ Industry_ 2026.pdf
- Music Business/ 02_ Revenue_ and_ Streaming/ 02_ WIPO_ Measuring_ IP_ Finance_ and_ Investment_ in_ Music_ 2026.pdf
- Corporate Finance/ 05_ Damodaran_ Valuation_ Approaches_ and_ Metrics_ 2006.pdf
- Corporate Finance/ 09_ Welch_ Present_ Value_ NPV_ 2022.pdf
- Financial Statement Analysis/ 20_ Penman_ What_ Matters_ in_ Company_ Valuation_ 2003.pdf

L18: Taxation + Music Business: Royalty Cash Systems

- Music Business/ 01_ IP_ and_ Rights/ 03_ USCO_ Copyright_ and_ the_ Music_ Marketplace.pdf
- Music Business/ 02_ Revenue_ and_ Streaming/ 05_ MMF_ Song_ Royalties_ Guide.pdf
- Taxation/ IRS_ Pub_ 334_ Tax_ Guide_ for_ Small_ Business.pdf
- Taxation/ IRS_ Pub_ 535_ Business_ Expenses_ Archived_ 2022.pdf
- Business Law/ 02_ Contracts/ sales- and- leases- ucc- article- 2- and- 2a- cali.pdf

L19: Market Demand as a Measurement System

- Marketing/ INDEX.md
- Marketing/ 01_ Foundations_ Purpose_ and_ Market_ Logic/ openstax_ principles_ of_ marketing.pdf
- Marketing/ 01_ Foundations_ Purpose_ and_ Market_ Logic/ intro_ consumer_ behaviour.pdf
- Marketing/ 02_ Market_ Understanding_ Data_ and_ Analytics/ marketing_ analytics_ methods_ practice.pdf
- Managerial Accounting/ 09 - CGMA - Global Management Accounting Principles.pdf

L20: Music Business + System Science: Audience Feedback Loops

- Music Business/ MANIFEST.md
- Music Business/ 03_ Audience_ and_ Fan_ Data/ 01_ MMF_ Fan_ Data_ Guide_ 2026.pdf
- Music Business/ 03_ Audience_ and_ Fan_ Data/ 05_ Arxiv_ Impact_ of_ Social_ Media_ on_ Music_ Demand.pdf
- System Science/ 01_ Foundations_ and_ Complex_ Systems/ networks_ crowds_ markets.pdf
- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ meadows_ leverage_ points.pdf

L21: Auditing + Marketing: Evidence for Growth Claims

- Marketing/ 02_ Market_ Understanding_ Data_ and_ Analytics/ marketing_ mix_ modeling_ bayesian.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ legal_ aspects_ marketing_ sales.pdf
- Auditing/ GAO_ Assessing_ Data_ Reliability_ 2019.pdf
- Auditing/ NIST_ SP_ 800- 137_ Continuous_ Monitoring.pdf
- Business Law/ 04_ Regulation_ Compliance/ ftc- doj- antitrust- guidelines- business- activities- affecting- workers- 2025.pdf

L22: Revenue Operations: From Funnel to Fulfillment

- Marketing/ 03_ Value_ Brand_ and_ Communication_ Systems/ emarketing_ 7e.pdf
- Marketing/ 04_ Acquisition_ Revenue_ and_ Governance/ power_ of_ selling.pdf
- Operations Management/ 21_ EPA_ Lean_ and_ Clean_ Value_ Stream_ Mapping.pdf
- Accounting Information Systems/ 13_ AsianInstitute_ Data_ Dashboarding_ in_ Accounting_ Tableau_ 2023.pdf
- Financial Accounting/ 22 - SEC - Beginners Guide to Financial Statements.pdf

L23: Integrated Profit Diagnosis

- Financial Accounting/ 01 - OpenStax - Principles of Accounting Volume 1 Financial Accounting.pdf
- Managerial Accounting/ 02 - Heisinger Hoyle - Managerial Accounting.pdf
- Cost Accounting/ 21_ IFAC_ Evaluating_ and_ Improving_ Costing_ in_ Organizations.pdf
- Financial Statement Analysis/ 08_ AICPA_ Analyzing_ Financial_ Ratios_ Practice_ Aid_ 2006.pdf
- Operations Management/ 20_ JIEM_ Theory_ of_ Constraints_ Make_ to_ Order_ Case_ Study.pdf

L24: Auditing + Operations: Control the Workflow

- Auditing/ GAO_ Green_ Book_ Internal_ Control_ 2025.pdf
- Auditing/ NIST_ SP_ 800- 53Ar5_ Assessing_ Security_ Privacy_ Controls.pdf
- Operations Management/ 14_ DiVA_ Digital_ Twin_ Bottleneck_ Diagnosis_ and_ Throughput_ Improvement.pdf
- Operations Management/ 24_ MDPI_ I3oT_ Sub_ Bottleneck_ Detection_ Continuous_ Improvement.pdf
- Accounting Information Systems/ 10_ MPRA_ Continuous_ Auditing_ Technology_ Review_ 2025.pdf

L25: Cash Flow vs Profit

- Financial Accounting/ 22 - SEC - Beginners Guide to Financial Statements.pdf
- Corporate Finance/ 17_ Welch_ From_ Financial_ Statements_ to_ Economic_ Cash_ Flows_ 2022.pdf
- Corporate Finance/ 23_ Welch_ Pro_ Forma_ Financial_ Statements_ 2022.pdf
- Taxation/ IRS_ Pub_ 538_ Accounting_ Periods_ and_ Methods.pdf
- Financial Statement Analysis/ 02_ Penman_ The_ Quality_ of_ Financial_ Statements_ 2003.pdf

L26: The Business Science Operating System

- System Science/ 02_ Dynamics_ Feedback_ and_ Leverage/ mit_ ocw_ sd_ checklist.pdf
- Accounting Information Systems/ 20_ WorldBank_ Financial_ Management_ Information_ System_ AIS_ ERP_ Dashboards.pdf
- Managerial Accounting/ 13 - Bourne Franco- Santos Micheli Pavlov - Performance Measurement System of Systems.pdf
- Business Law/ 04_ Regulation_ Compliance/ doj- evaluation- of- corporate- compliance- programs- 2024.pdf
- Music Business/ 04_ Market_ and_ Ecosystem_ Reports/ 01_ IFPI_ Global_ Music_ Report_ 2026_ State_ of_ the_ Industry.pdf

Final Capstone Project

The Business Science Operating System

Capstone objective: build a complete business system using all 13 subjects. The business may be a small operating company, a music venture, a service firm, or a productized advisory business.

Required components:

Business map: define customer, offer, operations, legal form, revenue streams, constraints, and feedback loops.

Accounting system: design transaction types, chart of accounts, source documents, reconciliation process, and dashboard outputs.

Financial statements: create pro forma income statement, balance sheet, and cash flow statement.

Cost structure: identify fixed costs, variable costs, activity drivers, bottlenecks, and unit economics.

Management dashboard: define metrics for demand, operations, cost, cash, tax, compliance, audit, and strategic progress.

Corporate finance decision: evaluate one investment using NPV, downside case, funding choice, and kill rule.

Tax considerations: identify entity treatment, income character, deductions, timing, payroll, withholding, and audit documentation.

Audit/control system: create controls over cash, revenue, vendor payments, payroll, contracts, marketing claims, and system access.

Legal risks: map contracts, IP, employment/contractor risk, advertising claims, governance, compliance, and liability exposure.

Marketing strategy: define target segment, positioning, channel system, funnel metrics, CLV/CAC logic, and test plan.

Music business application: apply the model to rights, royalties, streaming, fan data, catalog investment, or artist-service revenue.

Systems feedback loop: define weekly, monthly, and quarterly review cycles and show how evidence changes decisions.

Capstone execution steps:

1. Choose one business model and write the operating hypothesis.
2. Build the signal map: demand, activity, transactions, costs, risks, cash, controls, and market feedback.
3. Convert signals into meaning using accounting, ratio analysis, cost analysis, audit evidence, legal interpretation, and marketing analytics.
4. Convert meaning into value using profit, cash flow, tax savings, risk reduction, demand quality, and strategic advantage.
5. Execute one improvement experiment with a defined owner, metric, threshold, cost, and review date.
6. Update the business system based on measured feedback.

Final capstone deliverable: a Business Science Operating System binder with system map, AIS blueprint, statements, cost model, dashboard, capital decision, tax map, control checklist, legal risk register, marketing strategy, music-business application, and feedback review template.

Acceptance test: the capstone passes only if every major decision can be traced through signal -> meaning -> value -> execution and every major metric has an owner, source, threshold, and action rule.